

Exp: 2 - Write the Python Program to solve 8-Queen Problem

Aim:

To write a python program to solve 8-Queen problem.

Algorithm:

Step1: Start with an empty board of size 8x8.

Step2: Place the first queen in the first row and the first column.

Step3: For each subsequent row, check all possible columns to place the queen, and backtrack if a valid position cannot be found.

Step4: A valid position is a column where no other queens are in the same row, column, or diagonal.

Step5: If no solution is found, return failure.

Program:

```
N = 8
```

```
board = [["_." for _ in range(N)] for _ in range(N)]
```

```
def is_safe(row, col):
```

```
    for i in range(col):
```

```
        if board[row][i] == "Q":
```

```
            return False
```

```
    i = row
```

```
    j = col
```

```
    while i >= 0 and j >= 0:
```

```
        if board[i][j] == "Q":
```

```
            return False
```

```
    i -= 1
```

```
    j -= 1
```

```
    i = row
```

```
    j = col
```

```
    while i < N and j >= 0:
```

```
        if board[i][j] == "Q":
```

```
            return False
```

```

i += 1
j -= 1
return True

def solve(col):
    if col == N:
        return True
    for row in range(N):
        if is_safe(row, col):
            board[row][col] = "Q"
            if solve(col + 1):
                return True
            board[row][col] = "."
            return False
    if solve(0):
        print("Solution Found\n")
        for row in board:
            print(" ".join(row))
    else:
        print("No Solution")

```

Output:

Solution Found

```

Q . . . . . .
. . . . . Q
. . . Q . .
. . . . . Q
. Q . . . .
. . Q . . .
. . . Q . .
. . Q . . .
|

```

Result: Hence, the simple python program for 8- Queen problem is done